

Name of the student _____

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Indian Institute of Information Technology Bhopal

Department of Computer Science and Engineering / Information Technology

MINI Test – August 2025

Semester: 3rd**dr. Rahul chaurasiya**

Subject Code: CSE - 3003

Subject Name: Computer Organization and Architecture

Note:

1. Solve all five questions and while solving write question number and its part clearly.
2. All questions carry equal marks.

S. No.	Questions	Marks
1.	a) Consider MM of 512-Bytes and is divided into blocks of 16 words each. Whereas cache memory is of size 64-Bytes. Initially, cache is having 4th, 1st, 2nd, and 7th main memory blocks. Then determine whether the memory words 100 and 50 are present or not in the cache, if Associate Mapping is followed. <i>Assume 1 Byte = 1 Word.</i> b) Design the architecture for Direct memory mapping. Consider MM is of size 512-Bytes and cache memory is of size 64-Bytes. Each block contains 4 words. Determine the required hardware units and their size. (<i>Assume 1 Byte = 1 Word</i>) c) In a Direct mapped cache, there are N-TAG bits. The no. of TAG bits = No. of cache blocks. Each cache block contains N words. Determine the no. of words in Physical memory? d) Consider a cache memory with 80% Hit for Read operation and 90% HIT for write operation. The cache access time is 5 times faster than main memory access time, which is 20ms. If Write-back updation is used then if there is a Miss for read or write, a 2-word block (1 block = 1 word) must be moved from main memory to cache memory. Assume that cache is initially full and at any point of time 20% of its block have modified. Compute Avg. Read access time and Avg. write access time. e) Among the 1000 references in 64 words cache, the following observations are made. 100 references resulting Miss due to conflict problem, 200 references resulting Miss due to compulsory problem, and 100 due to capacity problem. What will be the HIT ratio if the cache is used with 4-way and 8-way set Associate mapping.	2X5 $\left\lfloor \frac{100}{16} \right\rfloor = 6$ $\left\lfloor \frac{50}{16} \right\rfloor = 3$ <i>Both misses</i> <i>Ans =</i> $N + 2 \log_2 N + 2^N \rightarrow 2 \log_2 N \rightarrow 2^N \rightarrow N^2$ $\log_2 N$ $0.675 \times 20 + 0.2 [2 \times T_{mm} + T_c]$ $16 + 40 + 4$

$$\frac{100}{4} = 25$$

$$\frac{100}{8} = 12.5$$

$$\frac{675}{1000} = 0.675$$

$$\frac{687.5}{1000} = 0.6875$$

$$0.8 \times 20 + 0.2 [2 \times T_{mm} + T_c]$$

$$16 + 40 + 4$$